About this PDF

Built only from: **Principles of Data Science** — https://openstax.org/books/principles-data-science/pages/1-5-data-science-with-python

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Covers 10 of 11 syllabus sub-topics of this unit.

Partly covered in the source:

- benefits and uses of data science
- facets of data
- setting the research goal
- exploratory data analysis methods
- presenting insights and building applications.

Not in the source (not covered in this PDF):

- data cleansing and data integration

Need for data science

Definition

Data science is a field that investigates how to collect, manage, and analyze data of all types to retrieve meaningful information and support decision-making. It integrates data collection, data management, and data analysis within a unified process.

Explanation

The proliferation of digital resources has produced massive amounts of data, making the analysis and modeling of this data essential for informed decisions in business, health care, education, and government. The data science cycle—problem definition, data collection, preparation, analysis, and reporting—provides a structured approach to turn raw data into actionable insights. However, the power of data analysis and modeling also raises ethical concerns, especially when data contain systematic bias or sensitive information. Addressing bias, ensuring fairness, and validating data through techniques such as cross-validation and outlier detection are crucial to maintain the integrity of the results.

Example

The COMPAS algorithm, used in the U.S. criminal justice system to predict reoffending risk, was found to flag Black defendants as high risk at nearly twice the rate of White defendants, leading to unjust sentencing and parole decisions. This case illustrates how biased data and models can produce unfair outcomes.

Algorithm / Steps

1. Define the problem statement to establish clear objectives.
2. Collect data systematically, ensuring relevance to the defined problem.
3. Prepare and process the data so it is suitable for analysis.

4. Analyze the data, applying statistical or machine learning methods.
5. Report findings and communicate insights.
6. Validate the model using techniques such as cross-validation and outlier detection to assess accuracy and fairness.

Key points

- Data science integrates collection, management, and analysis to extract meaningful information.
- The data science cycle includes problem definition, data collection, preparation, analysis, and reporting.
- Massive data generation necessitates rigorous validation and bias mitigation to support ethical decision-making.
- Bias in data can lead to unfair outcomes, as demonstrated by the COMPAS algorithm.

benefits and uses of data science

Partly covered in the source.

Definition

Data science benefits and uses involve the application of data visualization, data management, and cloud-based data warehousing to communicate information effectively and facilitate better decision-making.

Explanation

Data visualization is designed to present data graphically so that insights can be understood quickly, thereby improving decision-making. Data warehousing stores and manages large volumes of data from various sources in a central location, enabling efficient retrieval and analysis for business intelligence and decision-making.

Example

Enterprises subscribe to cloud-warehouse services such as Amazon RedShift or Google BigQuery instead of buying physical storage and configuring data management/processing systems on their own.

Key points

- Data visualization communicates information effectively and supports better decision-making.
- Data warehousing centralises large volumes of data, enabling efficient retrieval and analysis.
- Cloud-based data warehouses allow enterprises to avoid the costs of physical storage and maintenance.

facets of data

Partly covered in the source.

Definition

Facets of data refer to the different aspects that must be considered when working with data, such as how it is collected, prepared, validated, accessed, stored, and presented.

Explanation

These facets are important because data collection and preparation are often the most time-consuming steps in the data science cycle, and because accurate, ethical, and well-managed data is essential for reliable analysis and decision-making. Data validation, access management, and proper storage (e.g., cloud-warehousing) help ensure that data is trustworthy and usable.

Example

A smartphone automatically logs photos, videos, contacts, locations, and health data, which can be used for data collection without the need for domain experts to manually gather each type of information. Masking sensitive information, such as turning “Luke Smith, 1997, 567-89-1234, 123 Main Street” into “LS97, XXX-XX-XXXX, XXX Main Street,” demonstrates how data can be protected while still retaining useful identifiers.

Algorithm / Steps

1. Identify outliers that differ significantly from the rest of the data.
2. Perform cross-validation to assess the reliability of the data.
3. Apply ethical guidelines to ensure unbiased and responsible use of the data.

Key points

- Data collection can be automated (e.g., smartphone logs) or purposefully gathered by domain experts.
- Preparation must address issues such as language differences, time-zone variations, typos, and missing records.
- Validation involves outlier detection, cross-validation, and adherence to ethical standards.
- Modern data management uses cloud-based warehousing to store and process large volumes efficiently.
- Visualization and reporting are essential for communicating insights effectively.

overview of the data science process

The data science process is a structured approach to extracting insights from data.

Definition

Data science is a field of study that investigates how to collect, manage, and analyze data of all types in order to retrieve meaningful information.

Explanation

Data science tasks follow a process, called the data science cycle, which includes problem definition, data collection, preparation, analysis, and reporting. The cycle is designed to guide the work from defining objectives to communicating results. Data collection and preparation are often the most time-consuming steps, consuming about half of the total effort, while analysis and communication each take a quarter to a third of the time. Modern data scientists are expected to have expertise in all three domains—collection, management, and analysis—because technological advances have blurred the boundaries between them.

Example

A product sales and marketing team designs a customer satisfaction survey, collects responses, and later uses the data to analyze customer preferences. The survey data is gathered purposefully by domain experts and then processed for analysis.

Algorithm / Steps

1. Define the problem statement to establish clear objectives and scope of the project.
2. Generate and collect data on variables of interest.
3. Process the collected data so it is suitable for analysis (data preparation).
4. Analyze the prepared data to derive insights.
5. Report the findings to communicate results.

Key points

- The data science cycle consists of problem definition, data collection, preparation, analysis, and reporting.
- Data collection and preparation typically consume about half of the total project time.
- Modern data scientists need skills in data collection, management, and analysis due to blurred disciplinary boundaries.
- Example projects often involve purposeful data collection, such as customer surveys, followed by processing and analysis.

setting the research goal

Partly covered in the source.

Example

Mike decides to implement k-anonymization. To determine the appropriate value for k, Mike looks at the total number of respondents and the information collected in the survey. He knows that the more identifying characteristics he removes, the higher the level of anonymity will be. However, he also wants to make sure that the data is still useful for analysis. After some research, he determines that a minimum of 5 respondents should have the same characteristics in order to achieve a satisfactory level of anonymization. After reviewing the survey questions, Mike sees that there are 10 potential identifying characteristics, such as the number of children, their school district, and their ages.

Key points

- Data anonymization policy should be set by regulatory compliance officer or other compliance professionals with specialized knowledge of data privacy regulations.
- The policy must ensure that all possible security measures comply with prevailing data privacy regulations and that data masking processes use protected techniques.
- Mike determined that a minimum of 5 respondents should have the same characteristics to achieve satisfactory anonymization.
- The more identifying characteristics removed, the higher the level of anonymity, but data must remain useful for analysis.
- There are 10 potential identifying characteristics such as number of children, school district, and ages.

retrieving data from sources

Definition

Retrieving data from sources refers to accessing and loading data that is stored in various locations—such as local storage, cloud servers, or data warehouses—into an analysis environment for further processing.

Explanation

Cloud-based management systems store data on designated servers outside a local computer, allowing data scientists to retrieve and analyze large volumes of structured and unstructured data efficiently. Data warehousing systems centralise data from multiple sources, enabling quick access for business intelligence and decision-making. In a notebook environment, libraries like Pandas provide methods to load data files (e.g., CSV) into memory for analysis.

Example

A data scientist can create a folder on Google Drive, place a CSV file containing a dataset there, and then load that file into a Jupyter notebook using Pandas. Alternatively, enterprises may subscribe to cloud-warehouse services such as Amazon Redshift or Google BigQuery to store and retrieve data without managing physical storage.

Algorithm / Steps

1. Import the Pandas library and assign the abbreviation `pd` : `import pandas as pd`
2. Load the data file into a DataFrame, e.g., `df = pd.read_csv('path/to/file.csv')`

Key points

- Data can be stored locally or in cloud-based servers, and retrieval is performed from these sources.
- Data warehousing centralises data from multiple origins, facilitating efficient access.
- Pandas offers convenient functions to load data files into a notebook for analysis.
- Cloud-warehouse services like Amazon Redshift and Google BigQuery eliminate the need for physical storage.

data transformation techniques

Definition

Data transformation techniques modify raw data to improve its quality, privacy, or suitability for modeling. They include methods such as anonymization, outlier detection and handling, cross-validation, and k-anonymization.

Explanation

Anonymization removes or masks identifying information to protect privacy while preserving useful data. Outlier detection identifies abnormal values that may distort analysis; detected outliers can be removed or imputed to more reasonable values. Cross-validation evaluates a model's performance by repeatedly partitioning the data into training and testing sets, helping to detect over-fitting and bias. K-anonymization ensures that each combination of quasi-identifiers appears in at least k records, reducing the risk of re-identification.

Example

In a survey with 100 respondents, k-anonymization is achieved by setting $k = 5$, so each set of characteristics must contain at least 20 respondents. The survey replaces exact ages with age ranges and groups schools by district to meet this requirement.

Algorithm / Steps

1. **Identify sensitive attributes** that need masking or removal.
2. **Apply a masking technique** such as hashing or encryption to those attributes.
3. **Group data into equivalence classes** ensuring each class contains at least k records.
4. **Validate the anonymized data** by testing that no individual can be re-identified.
5. **Detect outliers** by statistical or algorithmic methods.

6. **Handle outliers**: either remove them or impute more reasonable values.
7. **Perform cross-validation**: repeatedly split the data into training and testing subsets, train the model, and evaluate performance on each subset.

Key points

- Anonymization protects privacy while retaining data utility.
- Outlier detection and handling prevent bias from abnormal values.
- Cross-validation assesses model robustness and detects over-fitting.
- K-anonymization requires a minimum of k records per equivalence class to reduce re-identification risk.

exploratory data analysis methods

Partly covered in the source.

Definition

Data analysis methods are used to discover meaningful insights.

Explanation

Most data scientists start with simple methods such as checking minimum and maximum values, and then move to more advanced methods like modelling a dependent variable, based on what they want to investigate further. Data validation techniques such as cross-validation and outlier detection help ensure the quality of the results.

Example

A concrete example of a simple exploratory method is checking the minimum and maximum values of a dataset.

Algorithm / Steps

1. Import pandas as pd.
2. Load the data into a DataFrame.
3. Use descriptive statistics like min() and max() to explore the data.
4. Apply data validation techniques such as cross-validation and outlier detection to refine the analysis.

Key points

- Data analysis starts with simple methods like checking min and max values.
- Advanced methods such as modelling a dependent variable are used after initial exploration.
- Data validation techniques like cross-validation and outlier detection help maintain quality.
- Pandas is a common library for data manipulation and analysis.

building predictive models

Definition

Building predictive models involves creating algorithms that use data to forecast outcomes, such as identifying patients at high risk of hospitalization or determining creditworthiness.

Explanation

The process requires careful data collection, bias assessment, training, validation, and ongoing monitoring to ensure fairness and accuracy.

Example

A health care startup developed a predictive model to identify patients at high risk of hospitalization due to chronic illnesses, collecting sensitive patient data but failing to implement robust data protection, leading to a breach.

Algorithm / Steps

1. Collect data and ensure quality and representation.
2. Split data into training and test groups.
3. Train the model on the training data.
4. Validate the model on the test data.
5. Monitor forecasted results and adjust as needed.
6. Perform human oversight by sampling large datasets for consistency.
7. Apply metrics to detect bias and ensure expectations are met.
8. Use a representative dataset to reduce bias, e.g., selecting diverse individuals.
9. Mitigate bias by implementing fairness techniques and evaluating algorithm design.
10. Employ interpretability and XAI to understand model decisions.

Key points

- Bias in data can lead to unfair or inaccurate predictions, necessitating regular bias checks.
- Fairness requires evaluating data quality, representation, training/validation procedures, and monitoring outcomes.
- Human oversight and metrics help detect and correct bias during model development.
- Representative datasets and feature selection are essential to reduce bias.
- Interpretability and XAI facilitate understanding and correcting biased decisions.

presenting insights and building applications.

Partly covered in the source.

Definition

Data reporting is the presentation of data in a way that best conveys the information learned from data analysis, often using visual elements such as charts, graphs, and maps to highlight patterns, trends, and hidden insights.

Explanation

Data scientists must communicate the results of their analysis to domain experts in a comprehensible manner; data visualization serves as a graphical method to present and report data, making it easier to understand and analyze.

Example

A data scientist might use a bar chart to show the distribution of customer satisfaction scores from a product survey, or a heat map to display search frequency by region.

Key points

- Data reporting conveys insights discovered during analysis.
- Visual elements like charts, graphs, and maps help communicate patterns and trends.
- Effective communication requires tailoring the presentation to domain experts.

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